Data Analyst Capstone Project

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OUTLINE



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EXECUTIVE SUMMARY



Technology trends from 65,437 developers, plus job-posting and salary data

- JavaScript, HTML/CSS, Python and SQL are the most-used languages today
- Python is the language developers most want to use next (44.9%)
  - Rust (30.9%) and Go (24.8%) enter the next-year top 10
  - Java and C drop out of the next-year top 10
- PostgreSQL leads databases now (50.8%) and next year (56.4%)
  - MySQL falls from 42.0% to 28.8%
- Job postings cluster in Washington DC, Detroit, Seattle and New York
- Swift, Python and C++ have the highest average salaries

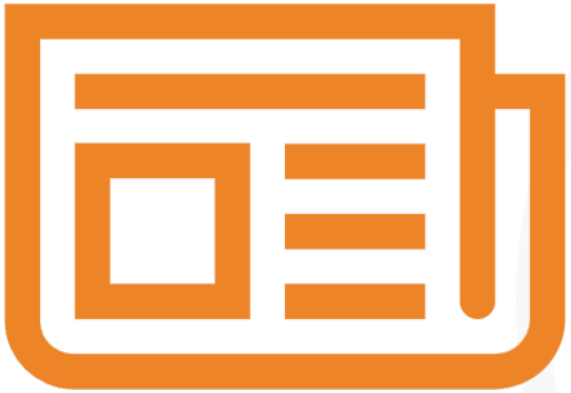
INTRODUCTION



Purpose of the report, target audience and value of the analysis

- Purpose: show which technologies developers use today and want to use next
- Audience: hiring managers, training teams and technology leaders
- Value: guides hiring, upskilling and tool-selection decisions
- Data: Stack Overflow Developer Survey, job postings and salaries
  - 65,437 survey respondents after cleaning
  - Job postings for seven US cities

METHODOLOGY



Data sources, data collection and data wrangling steps

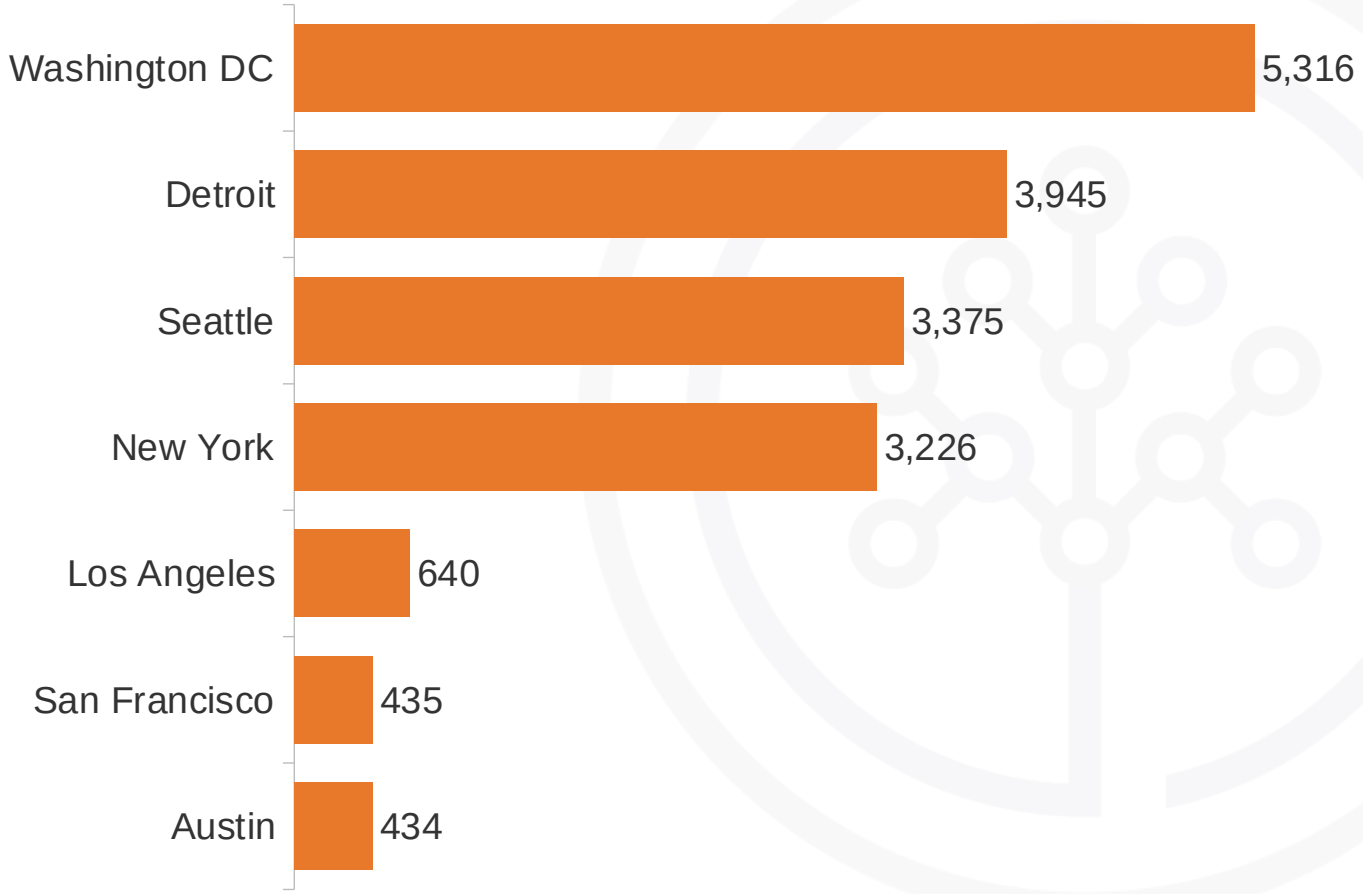
- Sources: survey CSV, job-postings API, web-scraped language salaries
- Collection: Python requests and BeautifulSoup, saved as job-postings.xlsx and popular-languages.csv
- Wrangling: removed duplicate responses, handled missing values, standardized labels
- Analysis: top-10 rankings, distributions and normalized compensation
  - Percentages are the share of respondents who answered each question
  - Multi-answer fields were split on semicolons before counting

Results



Job postings, popular languages, and language and database trends

JOB POSTINGS

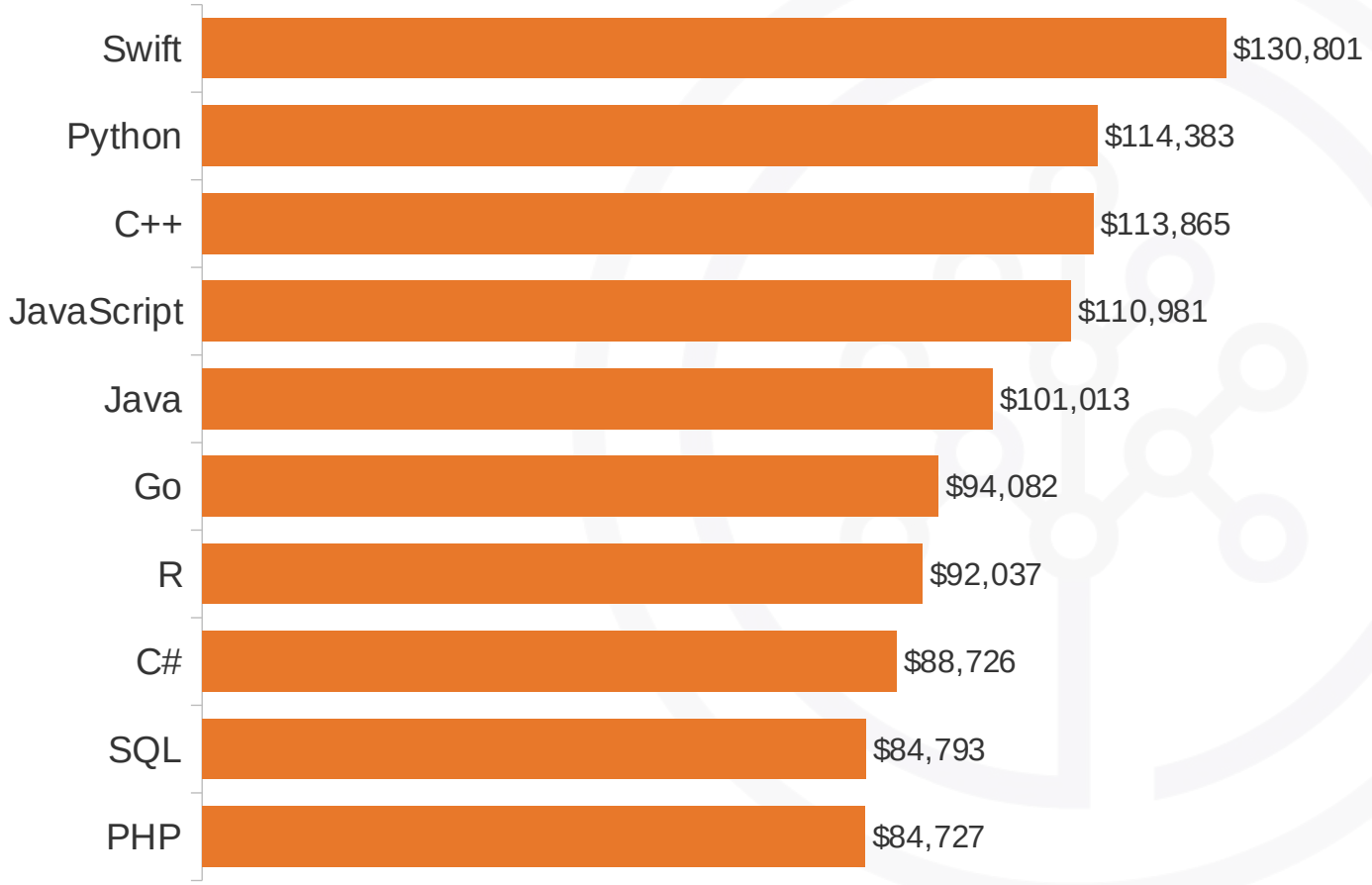


Washington DC has the most postings (5,316)

Detroit (3,945) and Seattle (3,375) follow

Los Angeles, San Francisco and Austin each have fewer than 650

POPULAR LANGUAGES



Swift pays the most ($130,801)

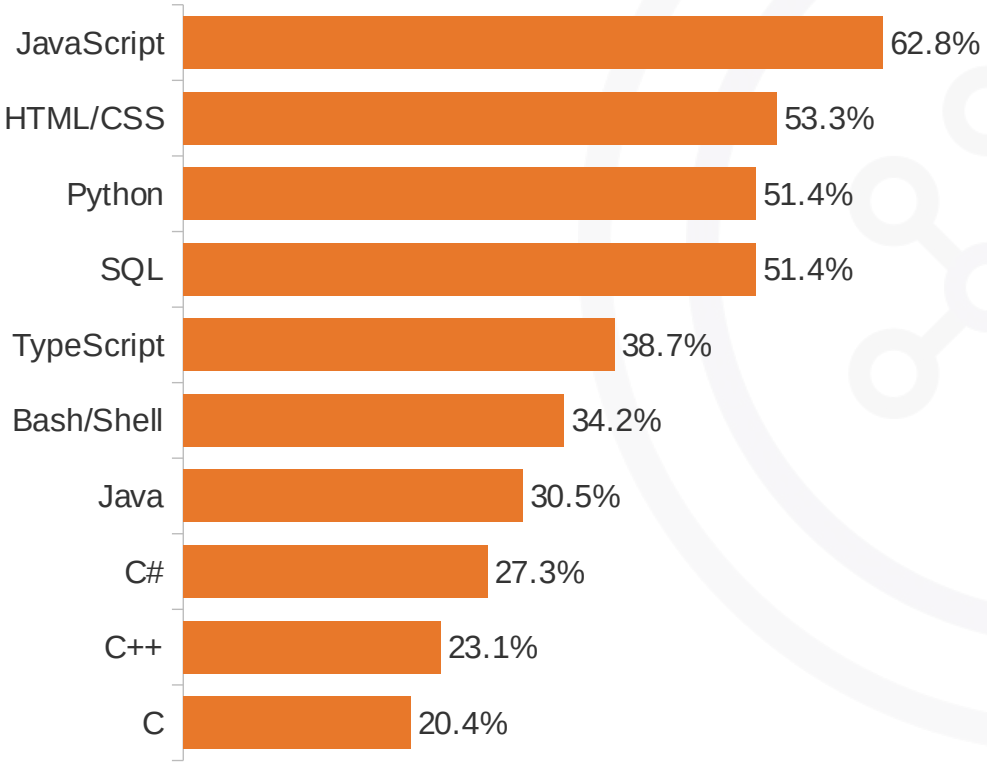
Python ($114,383) and C++ ($113,865) follow

SQL and PHP are lowest, near $85,000

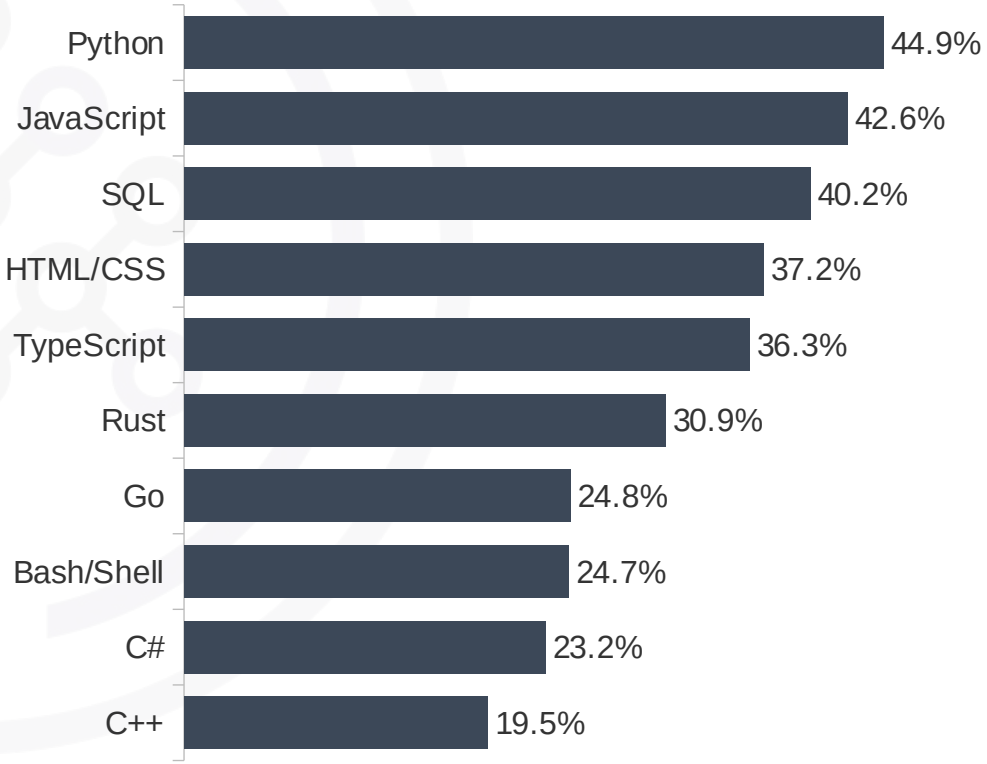
PROGRAMMING LANGUAGE TRENDS

Python is most wanted next year; Rust and Go enter the top 10

Current Year



Next Year



IBM

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript leads current use (62.8%); Python leads next year (44.9%)
- Rust (30.9%) and Go (24.8%) join the next-year top 10
- Java and C fall out of the next-year top 10

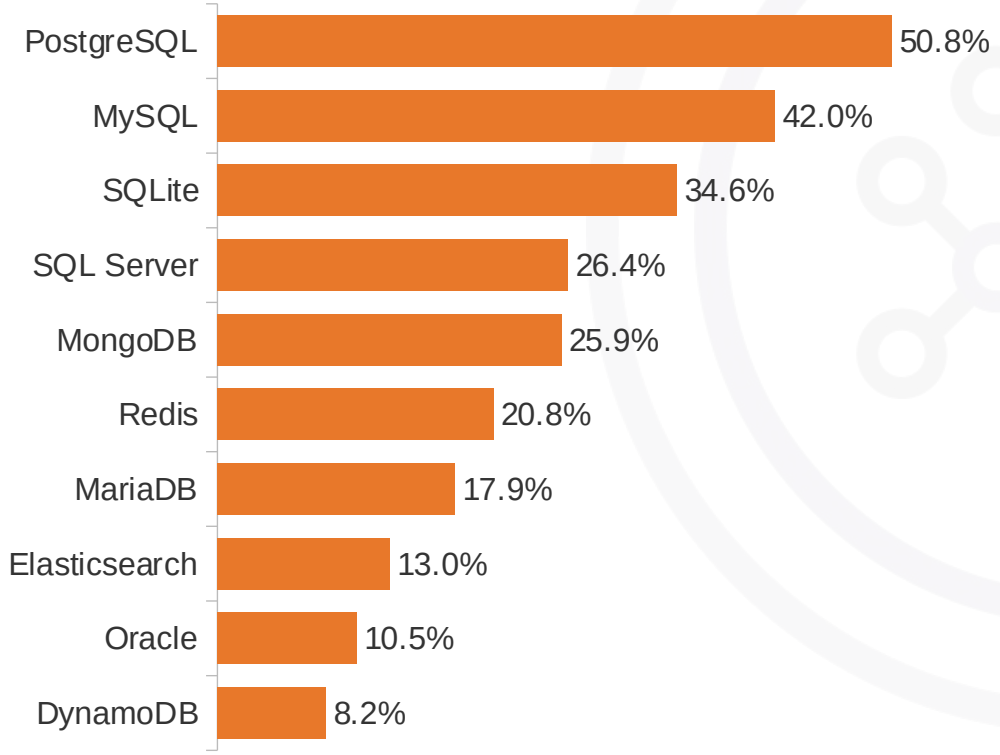
Implications

- Prioritize Python and SQL for data and general-purpose roles
- Treat Rust and Go as emerging skills to watch in hiring
- Keep JavaScript and TypeScript skills for web teams

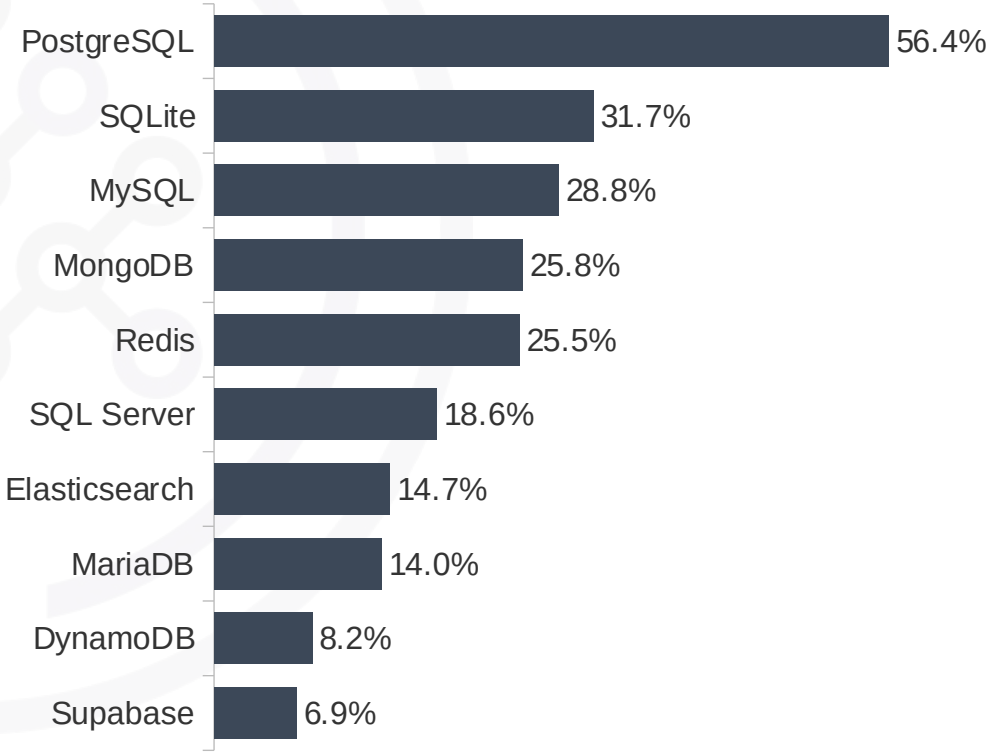
DATABASE TRENDS

PostgreSQL leads both years; MySQL declines, Supabase enters

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

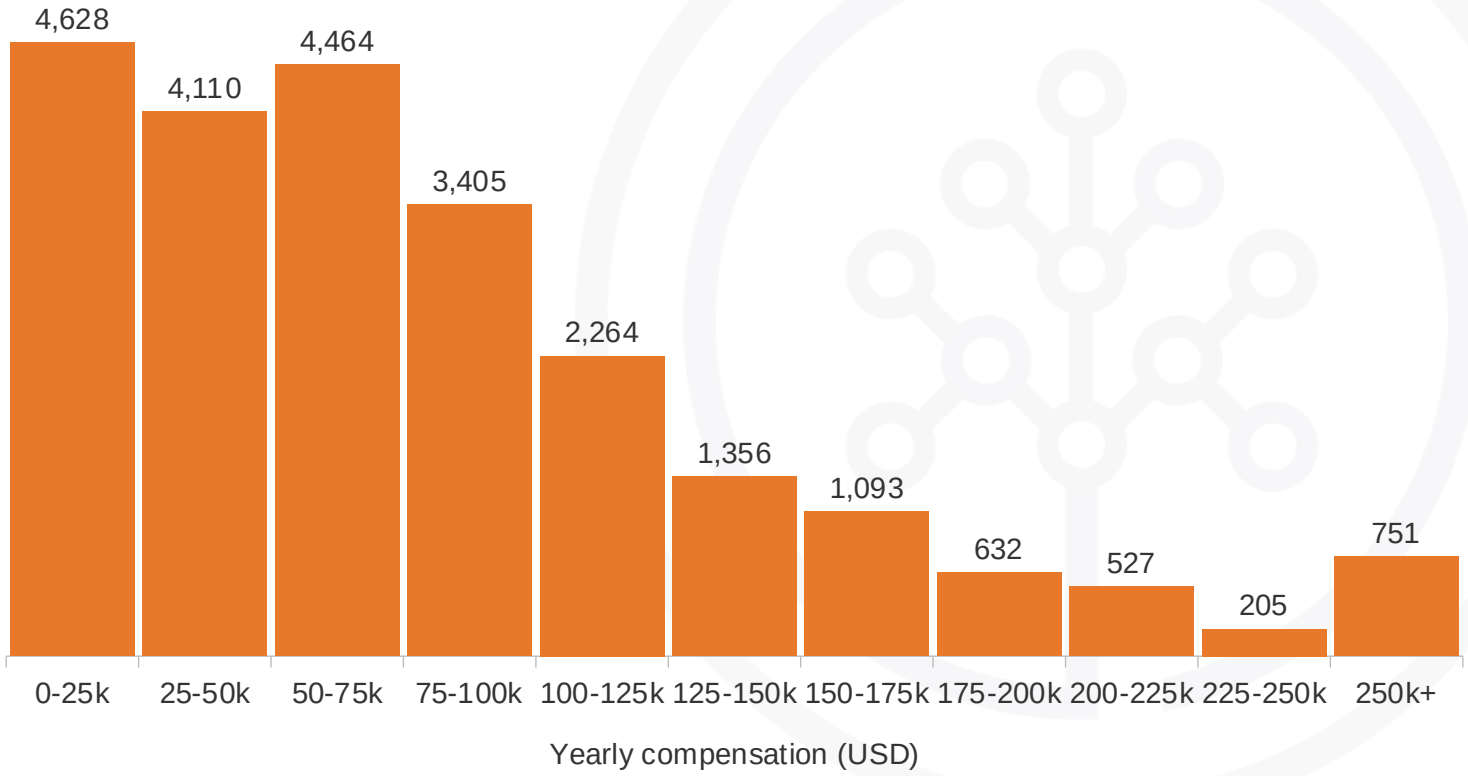
Findings

- PostgreSQL is first now (50.8%) and next year (56.4%)
- MySQL drops from 42.0% to 28.8%, behind SQLite (31.7%)
- Oracle leaves the top 10; Supabase (6.9%) enters

Implications

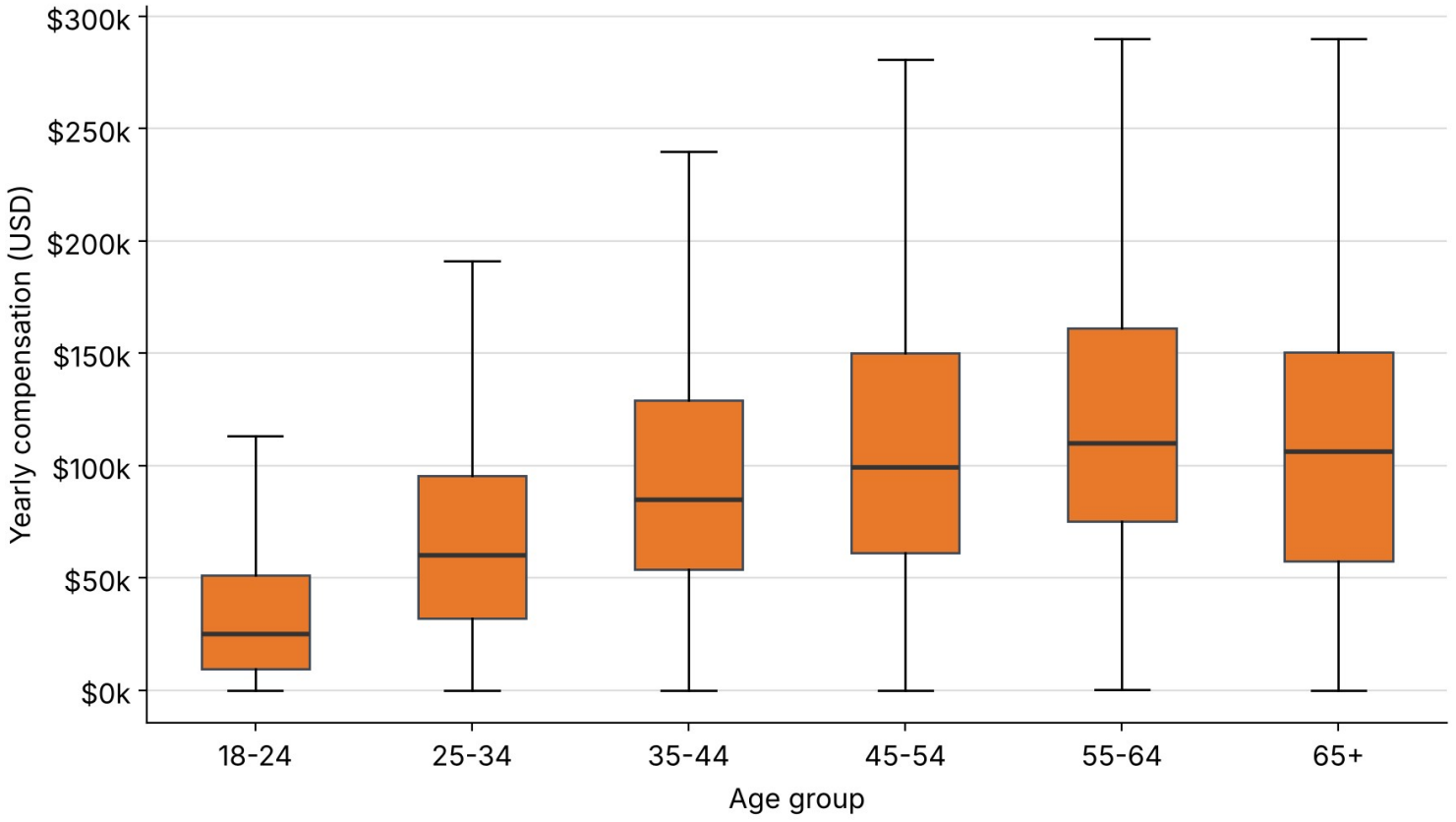
- PostgreSQL is the safest database skill to invest in
- Demand for MySQL and Oracle skills may soften
- Watch newer managed options such as Supabase

COMPENSATION DISTRIBUTION



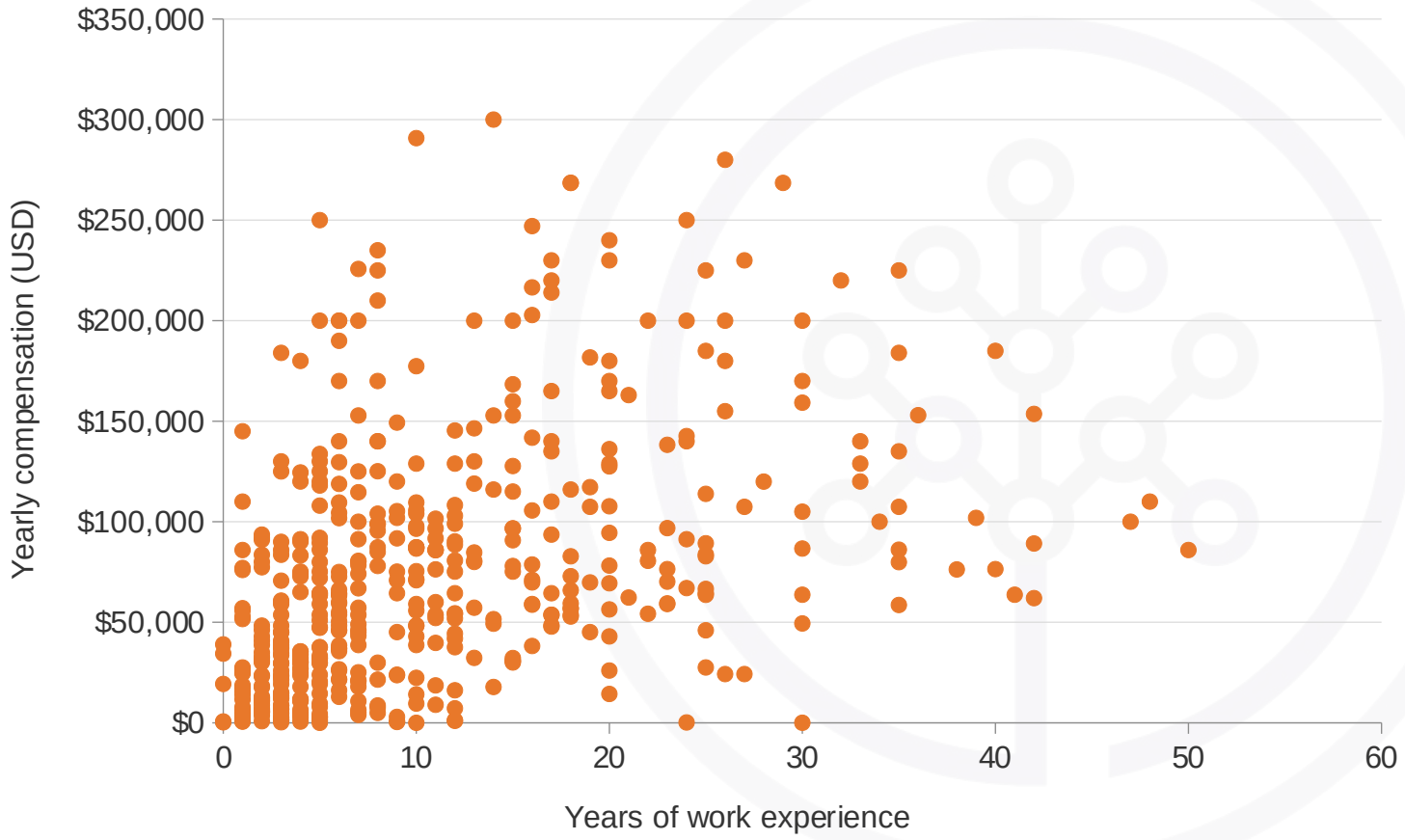
- Most respondents earn under $100,000 (71%)
- The most common bands are $0-25k and $50-75k
- 751 respondents earn $250,000 or more
- Based on 23,435 respondents with a reported salary

COMPENSATION BY AGE GROUP



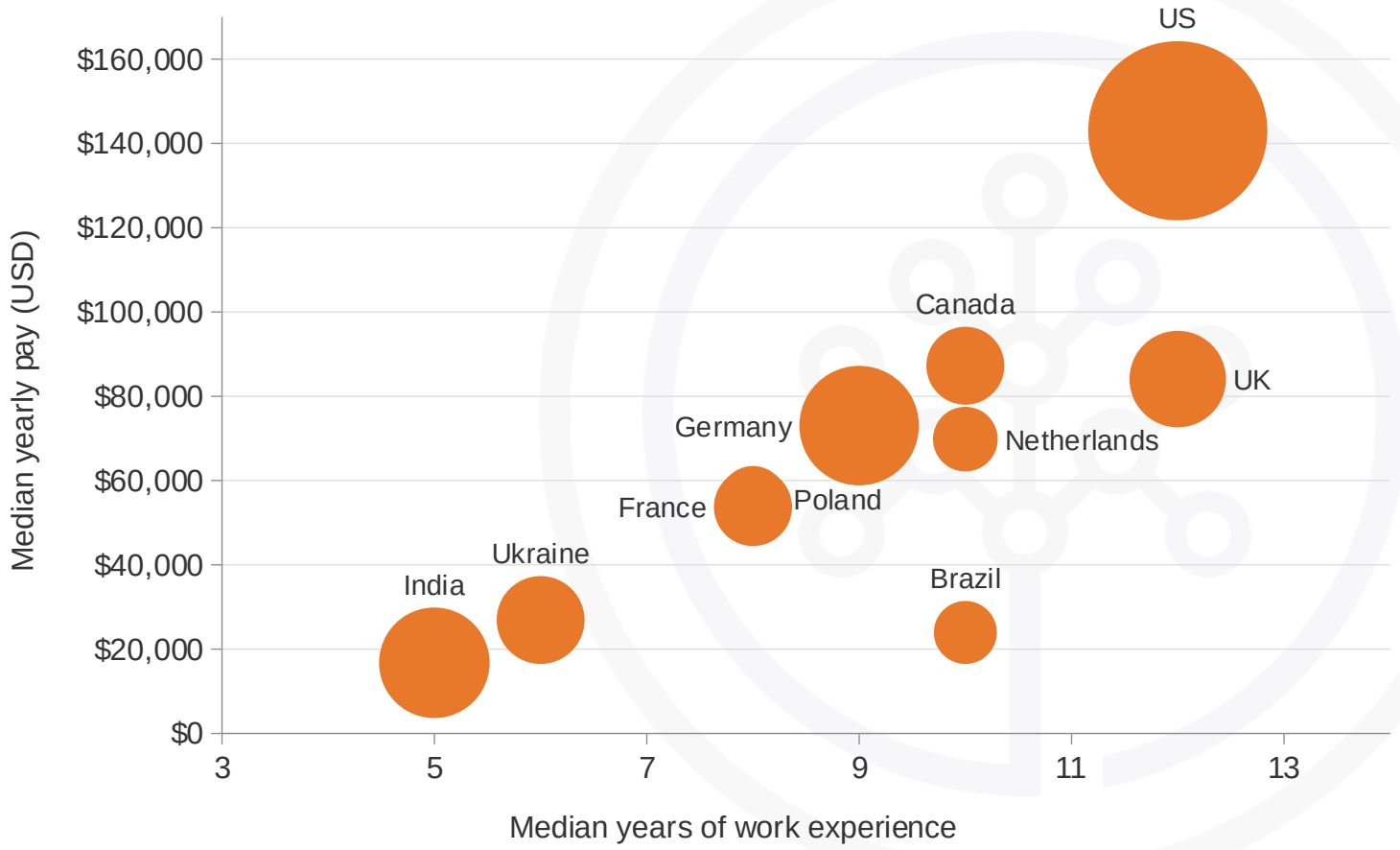
- Median pay rises with age, from $25,000 (18-24) to about $110,000 (55-64)
- The spread widens too: the middle half of ages 45-54 earns $61k to $150k
- Extreme outliers are hidden for readability

EXPERIENCE VS COMPENSATION



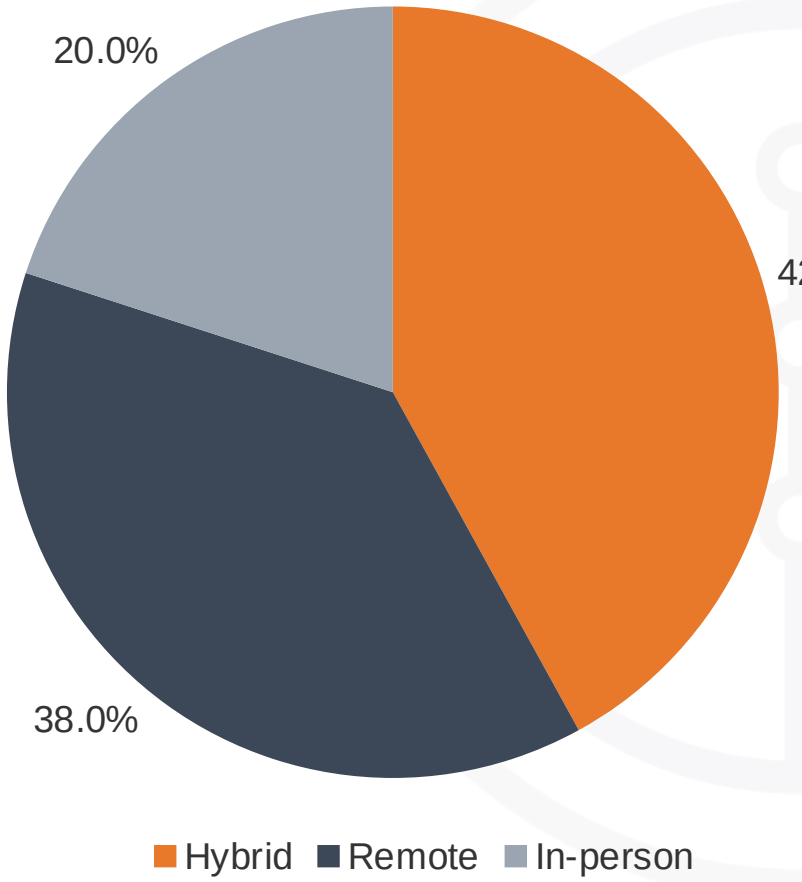
- Pay tends to rise with experience (correlation 0.41)
- The spread is wide, so experience explains only part of the pay gap
- 500 of 15,891 respondents shown; pay capped at $300,000

PAY AND EXPERIENCE BY COUNTRY



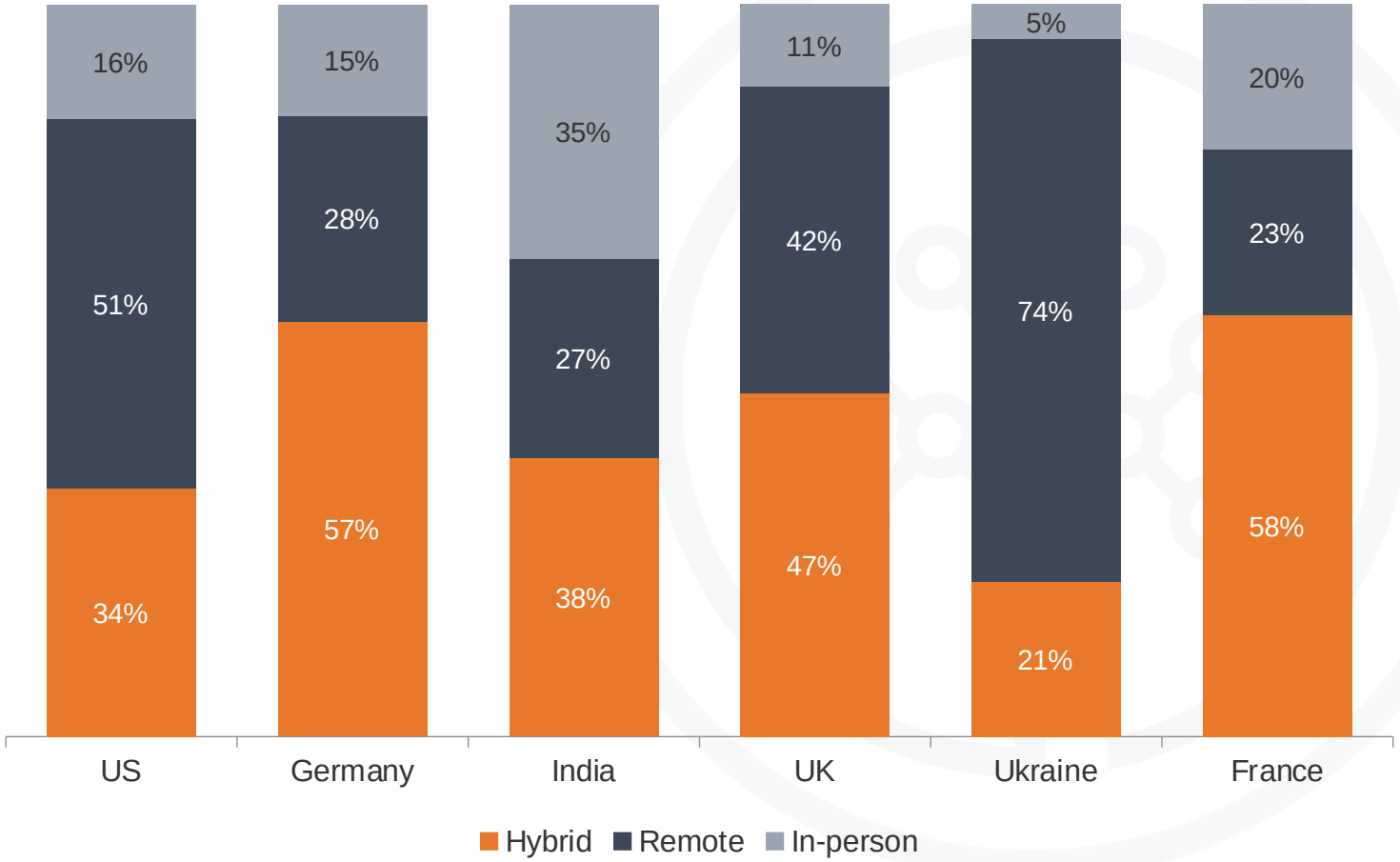
- The US has the most respondents and the highest median pay ($143,000)
- India ($16,749) and Ukraine ($26,904) have the lowest median pay
- Bubble size is the number of respondents; pay and experience are medians

REMOTE WORK SPLIT



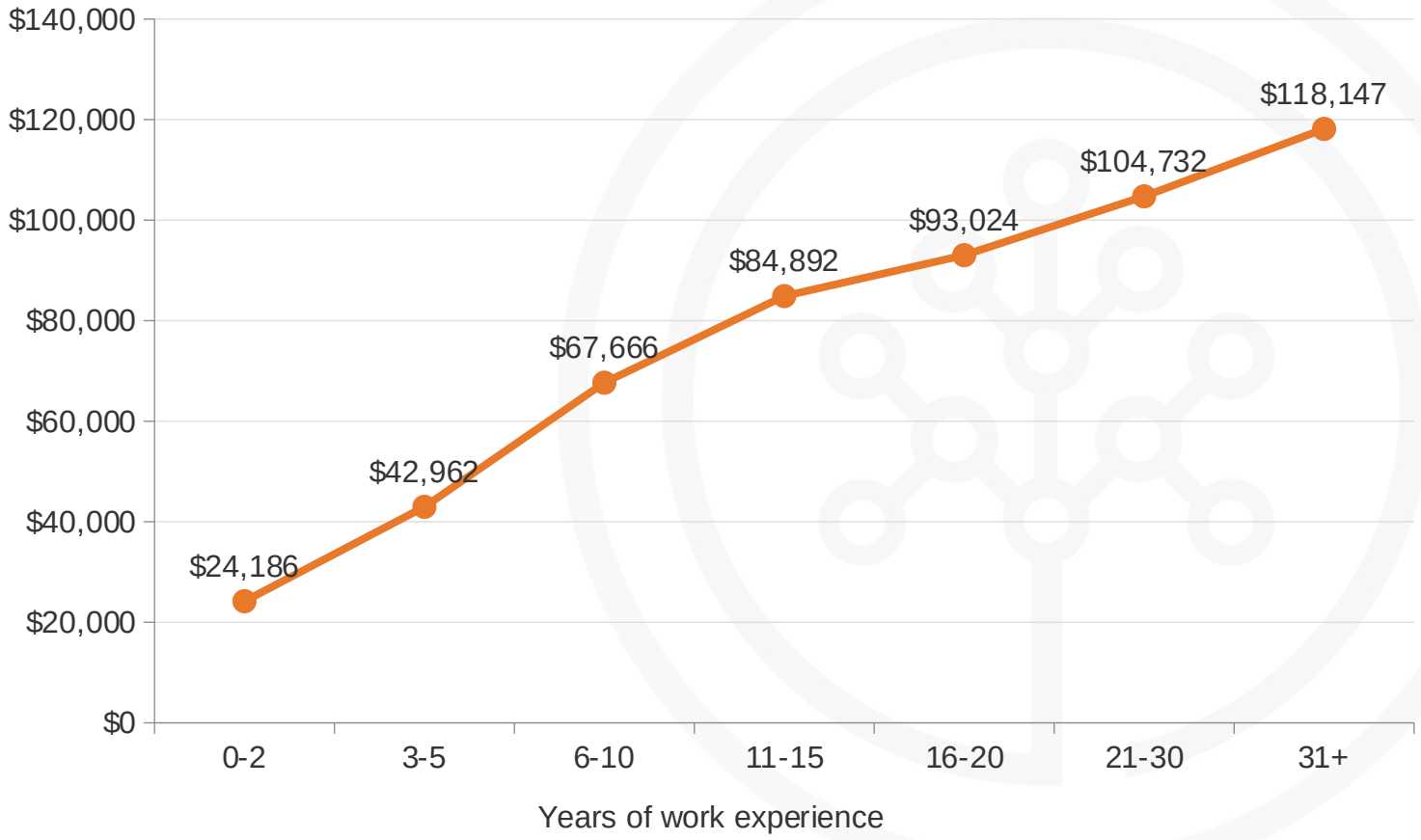
- Hybrid is the most common arrangement (42.0%)
- Only 1 in 5 respondents works fully in person (20.0%)
- Based on 54,806 respondents who answered

REMOTE WORK BY COUNTRY



- Ukraine is the most remote (74.2%), then the US (50.5%)
- France (57.6%) and Germany (56.7%) are mostly hybrid
- India has the highest in-person share (34.7%)

MEDIAN PAY BY YEARS OF EXPERIENCE



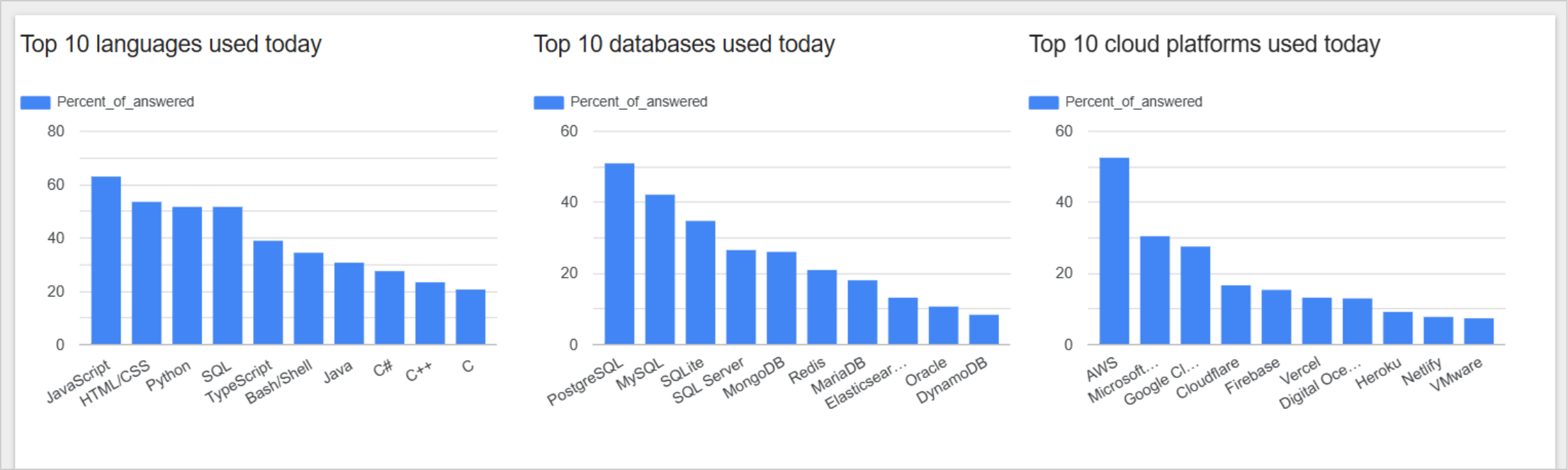
- Median pay rises with every experience band
- From $24,186 (0-2 years) to $118,147 (31+ years)
- Pay rises about 78% from 0-2 to 3-5 years

DASHBOARD



Three dashboard tabs: current technology usage, future technology trends and demographics

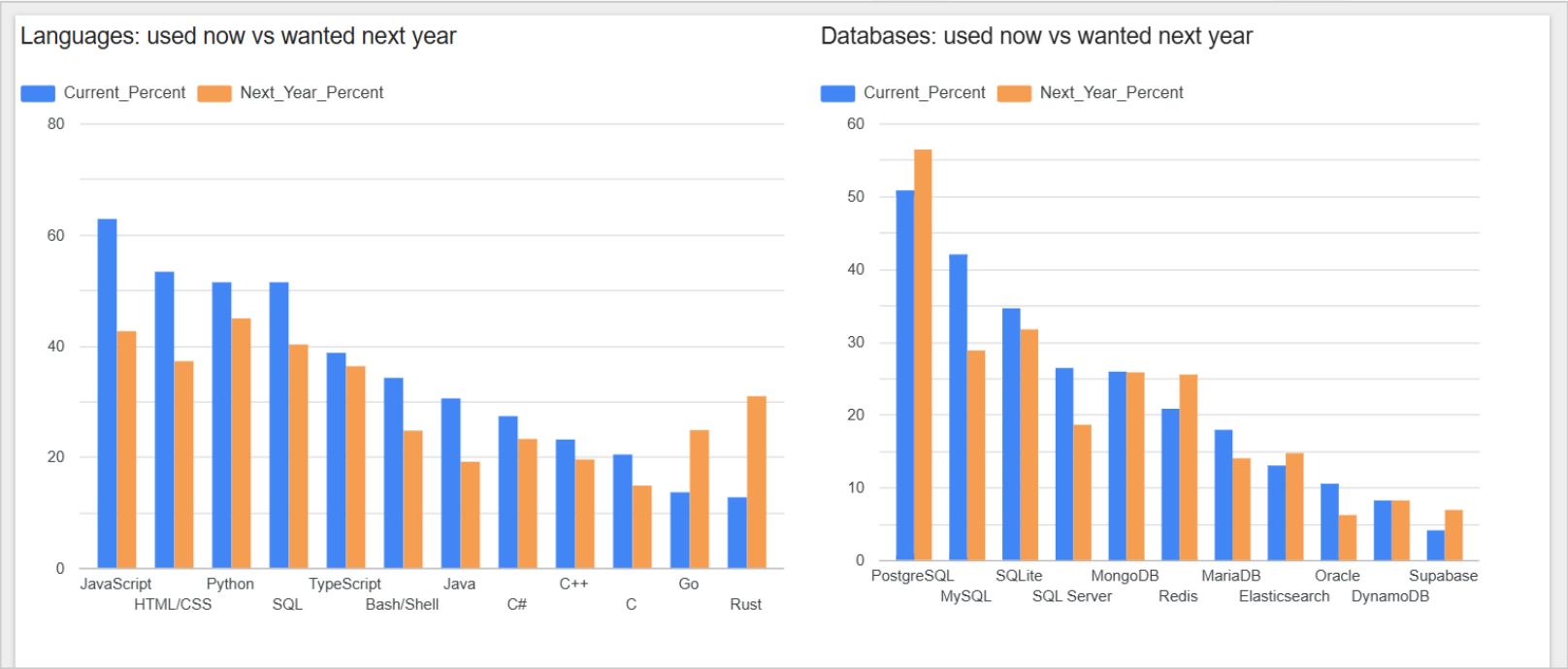
DASHBOARD TAB 1: Current Technology Usage



Dashboard Insight

- JavaScript (62.8%), HTML/CSS (53.3%), Python and SQL (51.4%) are the most used languages
- PostgreSQL (50.8%) and MySQL (42.0%) lead databases
- AWS is the top cloud platform (52.4%), ahead of Azure (30.3%) and Google Cloud (27.4%)

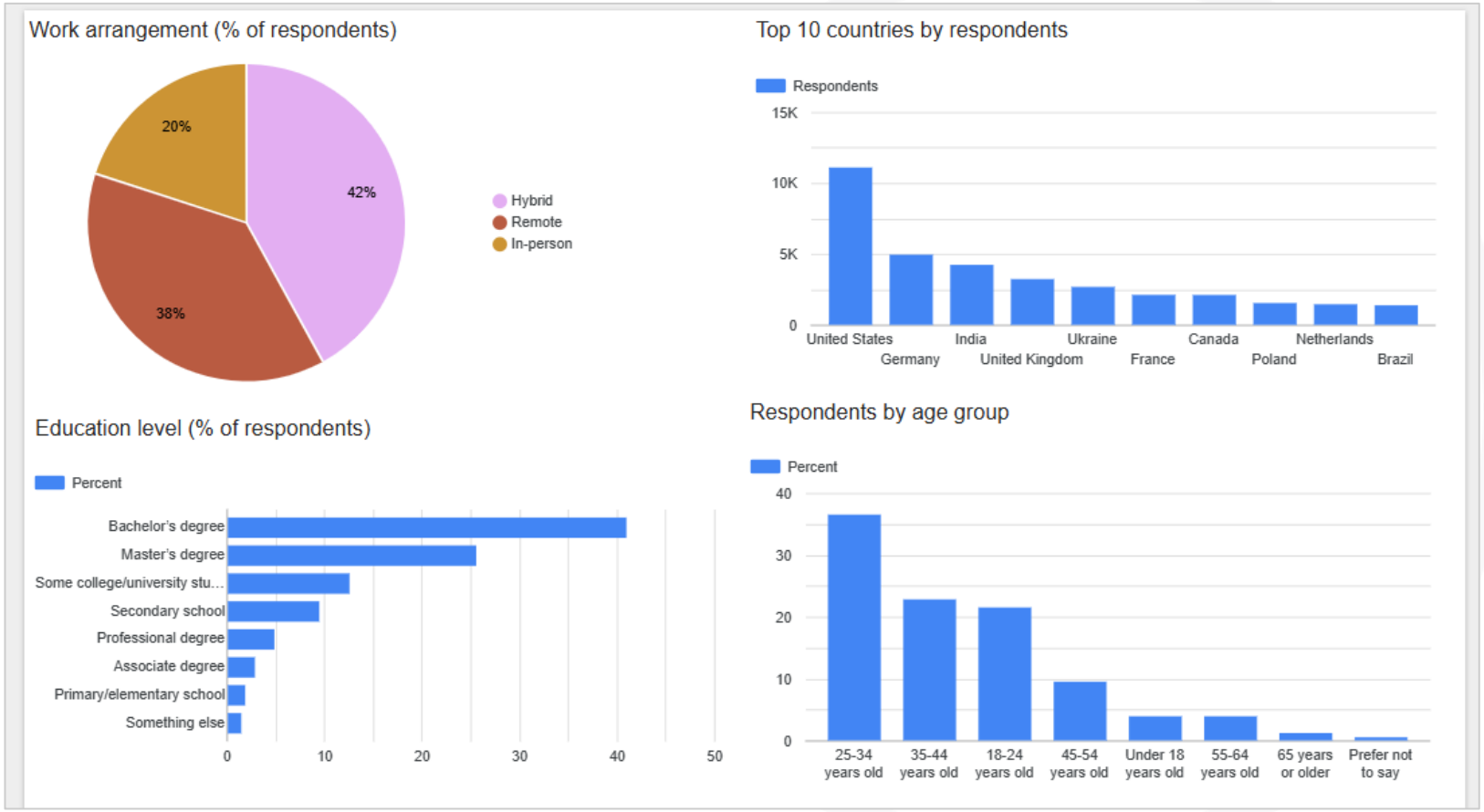
DASHBOARD TAB 2: Future Technology Trends



Dashboard Insight

- Python is the most wanted language next year (44.9%)
- Rust (12.7% to 30.9%) and Go (13.6% to 24.8%) show the biggest jumps
- PostgreSQL grows from 50.8% to 56.4%
- MySQL falls from 42.0% to 28.8%

DASHBOARD TAB 3: Demographics



Dashboard Insight

- 25-34 is the largest age group (36.5%)
- Hybrid (42%) and remote (38%) work dominate
- The US has the most respondents, then Germany and India
- Bachelor's (41%) and Master's (26%) are the most common degrees

DISCUSSION



- Python, JavaScript and SQL stay central; Rust and Go are rising
- PostgreSQL is the clear database favorite
- AWS leads cloud platforms; React and Node.js lead web frameworks
- Respondents skew 25-34 and mostly work hybrid or remote

OVERALL FINDINGS & IMPLICATIONS

Findings

- Python, JavaScript and SQL lead language use and interest
- PostgreSQL and AWS lead their categories
- Postings cluster in a few cities; Swift and Python pay most

Implications

- Focus training on Python, SQL and JavaScript
- Favor PostgreSQL and a major cloud platform
- Expect to pay a premium for Swift and Python skills

CONCLUSION



- Python and JavaScript remain core languages; Rust and Go are rising
- PostgreSQL is the dominant database now and next year
- Job postings and salaries vary widely by city and language
- Findings can guide hiring, training and technology choices

APPENDIX



- Additional tables from the analysis, on the next slide:
- Education levels of respondents
- Compensation summary (ConvertedCompYearly)

APPENDIX: SUPPORTING TABLES

Education level	Respondents	Share
Bachelor's degree	24,942	38.1%
Master's degree	15,557	23.8%
Some college, no degree	7,651	11.7%
Secondary school	5,793	8.9%
Professional degree	2,970	4.5%
Associate degree	1,793	2.7%
Primary school	1,146	1.8%
Something else	932	1.4%

Yearly pay (USD)	Value
Respondents with a salary	23,435
Median	$65,000
Mean	$86,155
99th percentile	$393,751
Maximum	$16,256,603

The mean is well above the median because a few very high salaries skew the distribution.